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A multilamellar organelle for chemosymbiosis in an aplacophoran mollusc adapted to anoxic cold seep sediment

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Symbiosis with chemoautotrophic bacteria has evolved in many animal lineages inhabiting reducing habitats such as hydrothermal vents, allowing these holobionts to thrive in dark biospheres. In certain cases, the symbionts have become intracellular, residing within specialised bacteriocytes. Although the integration of microbial symbionts with eukaryotic cells varies across known animals, no specialised organelle dedicated to chemosynthesis has been identified. Here, we report a mode of symbiosis where sulphur-oxidising bacteria cultured within spherical multilamellar compartments (~ 12 µm) in the cold-seep aplacophoran mollusc *Chaetoderma shenloong*. This organelle, which we name 'dracosphera', is ubiquitous within the hypertrophied and intricately reticulate digestive gland of *C. shenloong*, which has otherwise lost most of its gut. Given that the symbionts are microaerobic and the host resides in anoxic sediments tens of centimetres below the surface, the dracosphera may minimise oxygen diffusion to the bacteria, akin to mechanisms observed in microbial diazotrophs and termite hindguts, as supported by our genomic and spatial transcriptomic analyses. Hypoxia is known to induce radical adaptations in meiofauna, exemplified by the acquisition of hydrogenosomes. Our discovery of similar adaptations in *C. shenloong* provides insights into the evolution of such organelles in larger animals.

Symbiotic interactions are prevalent across the Tree of Life and span a spectrum of intimacy[1]. The success of eukaryotic life is largely attributed to the energy conferred by symbiosis, notably from the acquired organelles such as mitochondria, nitroplast, and chloroplasts[2,3]. A major scientific finding following the discovery of the first deep-sea hydrothermal vent on the Galápagos Rift in 1977[4] was that many animals inhabiting reducing habitats, such as hot vents and cold seeps, derive their energy and nutrients by 'feeding' on symbiotic bacteria that perform chemosynthesis using inorganic chemicals[5,6]. Today, we recognise chemosymbiosis as widespread throughout the ocean and

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underpins the high productivity of vent and seep ecosystems[7,8], with new and unexpected holobionts continuing to be discovered[9,10].

The mode of chemosymbiosis in large-bodied animals varies significantly. Some species cultivate and derive nutrition from epibiotic bacteria, as observed in various crustaceans[11,12]. Others occupy a transitional zone between extracellular and intracellular, where the symbionts reside within host vacuoles but maintain connections to the external environment, such as bathymodioline mussels and abyssochrysoid snails[13,14]. Additionally, some lineages exhibit complete encapsulation of symbionts by host bacteriocyte membranes, as seen in siboglinid tubeworms and peltospirid snails[5,6,10,15]. While the structures of symbiotic organs differ, most lack further compartmentalisation of symbionts within the bacteriocytes except siboglinid tubeworms[10,16–18] and lucinid bivalves[19], which both house them inside vacuoles. To date, no organelle dedicated to chemosynthesis has been identified.

Recently, a giant aplacophoran worm-mollusc from the understudied class Caudofoveata[20] was collected from black, reducing sediments of the Haima deep-sea cold seep in the South China Sea. This species, named *Chaetoderma shenloong* ('divine dragon' in Chinese)[21,22], exhibits a large size, reaching over 150 mm, in contrast to most species in the class, which average around 10 mm[23]. Despite its large size and habitat—buried approximately 40 cm below the sediment surface in black, reducing mud adjacent to a colony of symbiotic vesicomyid clams (Fig. 1a; Supplementary Fig. 1)—little is known about its ecology. The presence of multiple individuals within a single 6.5 cm diameter push-core in diameter (Supplementary Fig. 1 and Supplementary Movie 1) indicates that *C. shenloong* forms dense aggregations within the sediment, which becomes completely anoxic just 3 mm below the surface (Supplementary Fig. 1). This distribution is markedly different from typical caudofoveates, which usually burrow only in the upper 2–3 cm of the sediment[24]. Such high densities of large-bodied animals at chemosynthetic habitats are a hallmark of symbiotic species[25]. This raises the question: could *C. shenloong* harbour chemoautotrophic endosymbionts, and if so, where do the symbionts reside? Here, we address these questions using a variety of techniques, including traditional dissection, three-dimensional anatomical reconstruction with μ-computed tomography (*μ*-CT), fluorescent in situ hybridisation (FISH), electron microscopy, holobiont genomics, and spatial transcriptomics, revealing *C. shenloong* to be the animal with an organelle dedicated to chemosymbiosis.

Results and discussion

Anatomical and stable isotopic evidence

Although the external morphology of *C. shenloong* (Fig. 1b) does not diverge greatly from typical caudofoveates except for its large size and broad body[21], dissection of *C. shenloong* (Fig. 1c) revealed peculiar internal anatomy where most of the body cavity from the anterium throughout the trunk (demarcated by a constricted 'neck') is taken up by what seems to be a single coiled organ with haemocoel (blood) filling up the rest. Both 3D reconstruction from μ-CT scanning (Fig. 1d) and sectioning (Fig. 1e) showed this to be a complex, reticulated mesh-like organ. *Chaetoderma shenloong* has a mouth opening, and the buccal mass is sizeable (but only contains a very reduced radula)[21], which is then posteriorly divided into the oesophagus and the digestive gland as is typical in Caudofoveata (Fig. 1d). However, the oesophagus quickly narrows and we could only trace it for a short length before reaching the mid-point of the body. No hindgut or anus could be located in the posterior half of the body. As such, much of the digestive tract posterior to the oesophagus appears to have been lost in *C. shenloong*, which is present in typical omnivorous caudofoveates feeding on detritus, foraminifera, or small animals[24]. On the other hand, the digestive gland starts as an empty pouch but shortly fuses with the mesh-like symbiotic organ, which expands posteriorly to wrap around the gonad and also extends anteriorly to reach the buccal mass. Given its apparent connection with the digestive gland, this symbiotic organ is here interpreted to be a greatly hypertrophied and modified part of the digestive gland (diverticulum). Reduction or loss of the digestive tract is typical among chemosymbiotic holobionts[26]. The posterior is more typical of the class, containing a pericardium and a sizeable gill opening to the mantle cavity.

Our FISH results (Fig. 1e–g and Supplementary Fig. 2, including the probe details), combining general and symbiont-specific bacterial probes (EUB338 and Csh_SOB) revealed strongly localised chemosymbiotic bacterial signals in the digestive gland of *C. shenloong*, providing an initial indication that this organ is symbiotic. Using the digestive gland as a symbiotic organ has independently evolved in several molluscan lineages, most notably the extensive zooxanthellate tubular system in *Tridacna* giant clams for housing algal photosymbionts[27] and the 'solar-powered' slug *Elysia*, which uses digestive tubules to contain chloroplasts of a kleptoplastic origin[28,29]. This is also analogous to the symbiotic organs of the vent peltospirid snails *Chrysomallon* and *Gigantopelta*, which are modified oesophageal glands and therefore also originate from the digestive system[10,15]. Stable isotope ratios of carbon ($\delta^{13}C$) and nitrogen ($\delta^{15}N$) also lend support to *C. shenloong* relying on sulphur-oxidising symbiosis (Supplementary Fig. 3a), when compared to other fauna from the same seep[30]. While the $\delta^{13}C$ signature here indicates that the symbiosis is an important nutritional source, the extent to which symbiont-derived nutrition and energy support the host's overall energetic demand and maintaining its large body size requires future studies. Direct estimation of this energy contribution will require experiments on live animals, ideally integrating ^{13}C-bicarbonate incorporation rates with measurements of host oxygen consumption or respiration, or separating the symbiotic bacteria from the host and comparing the differences in $\delta^{13}C$ values between the host and symbiont. A close examination of the digestive gland of *C. shenloong* using confocal microscopy (Fig. 1f) showed that the symbiont signals (indicated by both EUB338 and Csh_SOB fluorescent probes) are restricted to small and spherical structures between 10-15 μm in diameter, and densely populating the reticulated diverticulum (Supplementary Fig. 2j–k).

A symbiotic architecture: the dracosphera

From scanning electron microscopy (SEM) of the digestive gland (Fig. 2a), we consider these spherical structures to be intracellular as they are wrapped inside a layer of membranous structure, most likely the cell membrane of the host's bacteriocyte. However, the complex nature of membranous structures within the symbiotic organ under transmission electron microscopy (TEM) makes the interpretation of cell boundaries challenging (Supplementary Fig. 4-5). These spherical structures are multilamellar and densely packed in bacterial cells under TEM (Fig. 2b), a mode of symbiont structural integration unlike any other known chemosymbiosis[8]. We here name this multilamellar organelle as the 'dracosphera'—a combination of *draco* meaning 'dragon' and *sphera* meaning 'ball, sphere' in Latin.

The dracospherae exhibited more than 20 layers of membranes (from 77 to 1060 nm, with an average value of 506 nm, Supplementary Fig. 6). This architecture represents a distinct evolutionary innovation compared to well-characterised chemosymbioses, including other molluscs and annelids[31–33]. Lucinid bivalves are notable for housing sulphur-oxidising symbionts in simple, single-membrane vacuoles (symbiosomes) within gill bacteriocytes[34]. This single-membrane structure is functionally analogous to a phagosome and primarily serves to contain the bacteria, facilitate nutrient exchange, and isolate bacterial metabolism from the host cytoplasm. The dracosphera, with its multilamellar structure, presents a far more complex and seemingly costly architecture, indicating a functional role beyond containment. Though multilamellar bodies have also been seen in the giant tubeworm *Riftia*, those were much smaller and are linked to symbiont digestion[35].

Our serial imaging of a complete dracosphera using focused-ion beam scanning electron microscopy (FIB-SEM) combined with FISH

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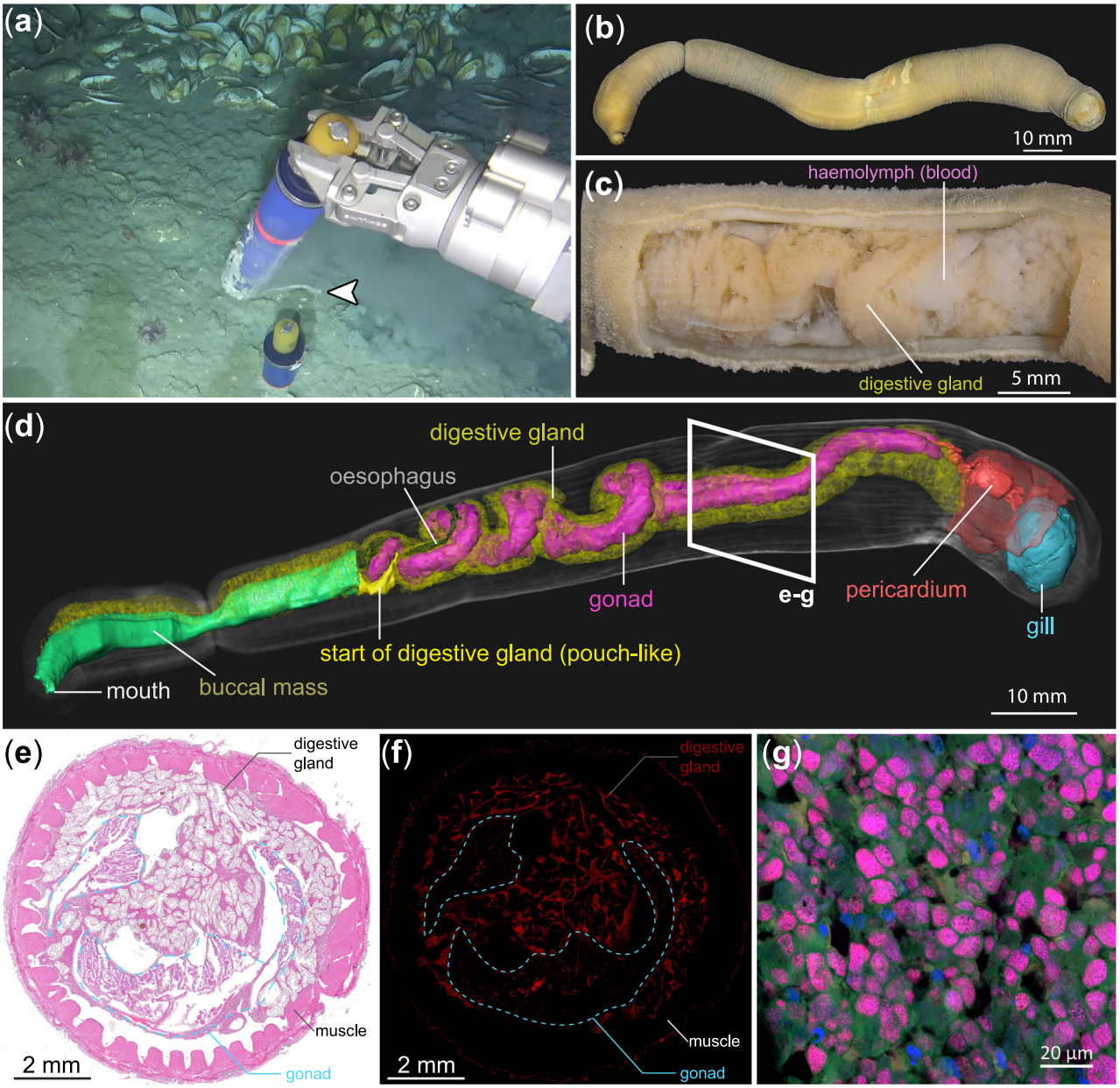


Fig. 1 | General information for *Chaetoderma shenloong* about habitat, morphology, and chemosymbiont signals in tissue. a In situ image of *Chaetoderma shenloong* while being sampled from deep sediments around a colony of vesicomyid clams using a pushcorer (Supplementary Movie 1). **b** External morphology of *C. shenloong*. **c** The trunk with the body wall dissected away to show the symbiotic organ (digestive gland). **d** 3D reconstruction of the key internal organs from µ-CT scan. **e** Histological section stained with hematoxylin and eosin. **f** Spatial distribution of symbionts in the section shown by fluorescence in situ hybridization (FISH) of a symbiont-specific probe (red signal). **g** A close-up FISH imagery localising symbionts within spherical structures ('dracosphera'); showing symbiont (red), nucleus (blue), membrane (green), and eubacteria (cyan). The blue dashed line indicates the outline of the gonad, showing some symbiont signals observed in the gonad, in addition to those in the digestive gland.

confirmed the absence of eukaryotic nucleus and lysosomes within the dracosphera but small vacuoles with a single layer of membrane were present (Supplementary Fig. 4 and Supplementary Movie 2). Compared to the common "farming" strategies via phagosome digestion in other chemosymbiotic molluscs[32], the large size of dracospherae suggests that host control is not exerted through responsive digestion upon overcrowding but rather through a relatively long-term maintenance.

Molecular characterisation of the symbiosis

A single sulphur-oxidising bacterium (SOB) symbiont species in the genus *Candidatus* Thiodubiliella was found inside the dracosphera, up to 1629 symbiont cells per eukaryotic cell within the reticulated mesh-like digestive diverticula (Supplementary Fig. 7d). It encodes only 1022 genes in a 1.14 Mb genome that is more reduced than its close relatives (*Bathymodiolus* and *Conchocele bisecta* SOB symbionts, Supplementary Fig. 7a-b, and Supplementary Data 2). Three genes are absent in the TCA cycle and two in the biosynthesis of amino acids (Methionine and Tyrosine), which are compensated by the host (Supplementary Fig. 8 and Supplementary Data 3). Remarkably, the genomic and transcriptomic analyses showed that the symbiont is capable of microaerobic or even facultatively anaerobic life, actively expressing genes in dissimilatory nitrate reduction that likely play vital roles as the electron acceptor (Supplementary Fig. 8-9). The loss of the caa_3-type cytochrome oxidase[36] (Supplementary Data 4), combined with the

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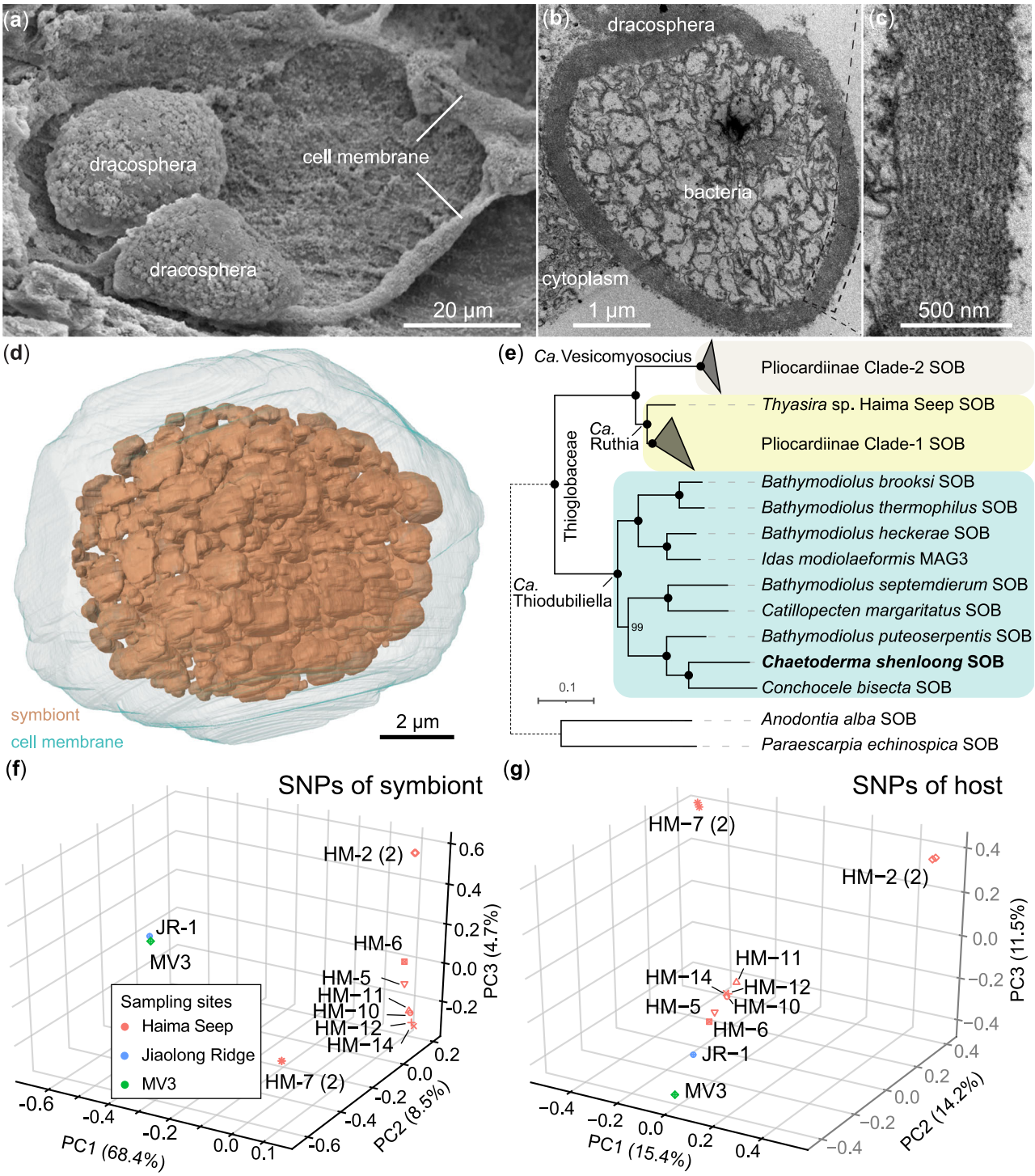


Fig. 2 | Ultrastructural morphology, 3D architecture, and genomic characteristics of dracospherae and the *Chaetoderma shenloong* symbiont. a Scanning electron micrograph of two dracospherae enveloped by a membrane. Two dracospherae were inside the cell membrane. **b** Transmission electron micrograph showing cross-section view of a dracosphera. Images shown in panels **a** and **b** are representative images from three independently prepared specimens, all yielding similar results. **c** Close-up of the multilamellar wall of the same dracosphera shown in **b**. **d** 3D reconstruction of a dracosphera from FIB-SEM (focused ion beam scanning electron microscopy) data showing tightly-packed symbiont cells completely enveloped. Light blue in semi-transparency represents the dracosphera membranes, while the symbiont cells inside are shown in brown. **e** Phylogenetic position of the *Chaetoderma shenloong* symbiont (IQ-TREE 2, best-fitting model with partition). **f** Principal component analysis (PCA) of symbiont single-nucleotide polymorphisms (SNPs) plotted using the top three components. **g** The same for host SNPs.

presence of the cbb_3-type with a high affinity for oxygen under hypoxia, indicates this SOB is adapted to hypoxic environments[37–39].

Chemosymbiotic clams such as pliocardiines and thyasirids, as well as many siboglinid tubeworms also obtain hydrogen sulphide from deep within the sediment by uptake through the foot or root, respectively[40]. However, their respiratory organs (e.g., the gill of bivalves and the plume of siboglinid tubeworms) are constantly exposed to oxic seawater, while the gill of *C. shenloong* would face a

much lower oxygen level (at least hypoxic or even anoxic) when burrowing deep to access hydrogen sulphide. When oxygen supply runs low, it would presumably then move up the burrow and allow the gill to reach the seawater layer above the sediment to replenish oxygen. Casting of their burrows and in situ measurements of the physico-chemical parameters of these burrows[41] will both be useful in gaining further insights on the ecology and behaviour of *C. shenloong*.

We examined the genetic divergence patterns of both the host and symbiont using single nucleotide polymorphisms (SNPs), where a congruent pattern suggests vertical transmission while a conflicting pattern would indicate horizontal transmission. Though these specimens were from three different seeps, we detected a clear signal of strong genealogical connectivity supporting them belonging to the same species, with more than 98.79% and 98.89% average nucleotide index (ANI) at the mitochondria and the whole-genome level, respectively (Supplementary Fig. 10). The identical symbiont SNPs observed in the anterior and posterior digestive gland of the same host individual (Fig. 2g and Supplementary Fig. 7c) suggests a limited time window for the symbiont acquisition. Among SNPs from eight hosts collected from Haima (Fig. 2g), two individuals ('HM-2', 'HM-7') were divergent from the rest, which also had symbiont SNPs that differed from the others. We also detected some symbiont signals in the gonad through FISH (Fig. 1f). Overall, these lend support to a vertical transmission of the symbionts, which is consistent with the very small genome size of the symbiont. Nevertheless, a mixed mode of symbiont acquisition like solemyid clams[42] and *Chrysomallon* snails[43] cannot be ruled out.

Host and symbiont interactions

As a reference, we assembled a chromosome-level genome of *C. shenloong* from Haima (Supplementary Tables 8-9), updating from an existing one collected from Jiaolong Ridge, also in the South China Sea[22]. The co-localisation of symbiotic reads and an unsupervised cluster of host reads highlights the potential to investigate the host and symbiont interaction, indicated by the spatial (meta)transcriptomics (Fig. 3a–c and Supplementary Fig. 11). Under the aggregated bin 20 (20 ×20 DNA Nano Balls, i.e. 10 µm diameter), the average numbers of the captured molecular identifiers (MID) and genes per bin in 5 sections were 56 and 23, respectively (Supplementary Table 7). There was a significantly higher ratio of symbiont-derived reads in symbiotic clusters compared to the rest, up to 45% vs. 27% in chip-5 ($p < 0.0001$, Fig. 3d). In total, 253 candidate genes were selected with the threshold as marker genes across at least two independent sections (Fig. 3e, labelled in bold, details shown in Supplementary Data 5) and then further classified on their functions (Fig. 3f, g and Supplementary Data 6-7).

Spatial transcriptomics revealed an enriched pattern of proteolysis-related genes in the digestive gland, including serine-type peptidase and metallopeptidase activities (Fig. 3f). These could aid the digestion of symbiont proteins as a source of amino acids[44], which would complement the deficiency of the host's biosynthesis capacity (Supplementary Fig. 9 and Supplementary Data 3). We did not observe lysosomes in dracospherae, but densely packed lysosomes (~ 200 nm diameter) were present in the adjacent cytoplasm (Fig. 2 and Supplementary Fig. 4b-c), and we also found evidence of hexosaminidase (enzyme in lysosome, Supplementary Data 5) in the symbiotic region. There were higher expression levels of genes in phagocytosis in the digestive gland compared to other tissues, evidenced by bulk RNA-seq (Supplementary Fig. 12).

It is hypothesised that the multi-membranes of the dracosphera prevent lysosomes from entering the dracosphera and the enormous size of the dracosphera also hinders direct digestion via traditional phagocytosis, unlike other chemosymbiotic holobionts. Therefore, we hypothesise that the dracosphera may exhibit controlled membrane disintegration at some point to release the symbionts into the cytoplasm—which can then be digested by the lysosomes (Fig. 4). There seems to be some capacity for the host to regulate the apoptosis or phagocytosis when symbionts are released from dracospherae, indicated by higher expression levels of peptidase inhibitors in the digestive gland (22 of 63 genes in the genome, Supplementary Fig. 13). Such a disintegration might be regulated by the host, such as by using specific enzymes, or by the symbionts physically exceeding the carrying capacity of dracospherae after divisions—requiring more in-depth studies in the future. In some TEM micrographs (Supplementary Fig. 4), the bacteria within the dracospherae differ in electron density with some appearing much darker than others, which may be related to the sulphur, glycogen, and polyhydroxyalkanoate (PHA) storage within the symbiont cells[45,46]. Although we did not see any structures that could be identified as lysosomes within the dracospherae, we cannot exclude completely an alternative mechanism that the dark and light bacteria represent intact ones and those undergoing digestion within the dracosphera, respectively. While our evidence points toward a "farming" strategy where the host utilises nutrients via symbiont digestion, the potential for a symbiont "milking" model where the symbionts translocate nutrients to the host through the dracospherae cannot be excluded entirely. Under the milking model, symbionts could actively transfer organic metabolites to the host potentially utilising some micro-pores (not identified in this study) in the membrane serving as specialised interfaces for nutrient exchange. It is possible that the host adopts a combination of milking and digestion to maximise energy acquisition, a hypothesis that could be tested via in vivo isotopic tracing in living specimens in the future[31,47].

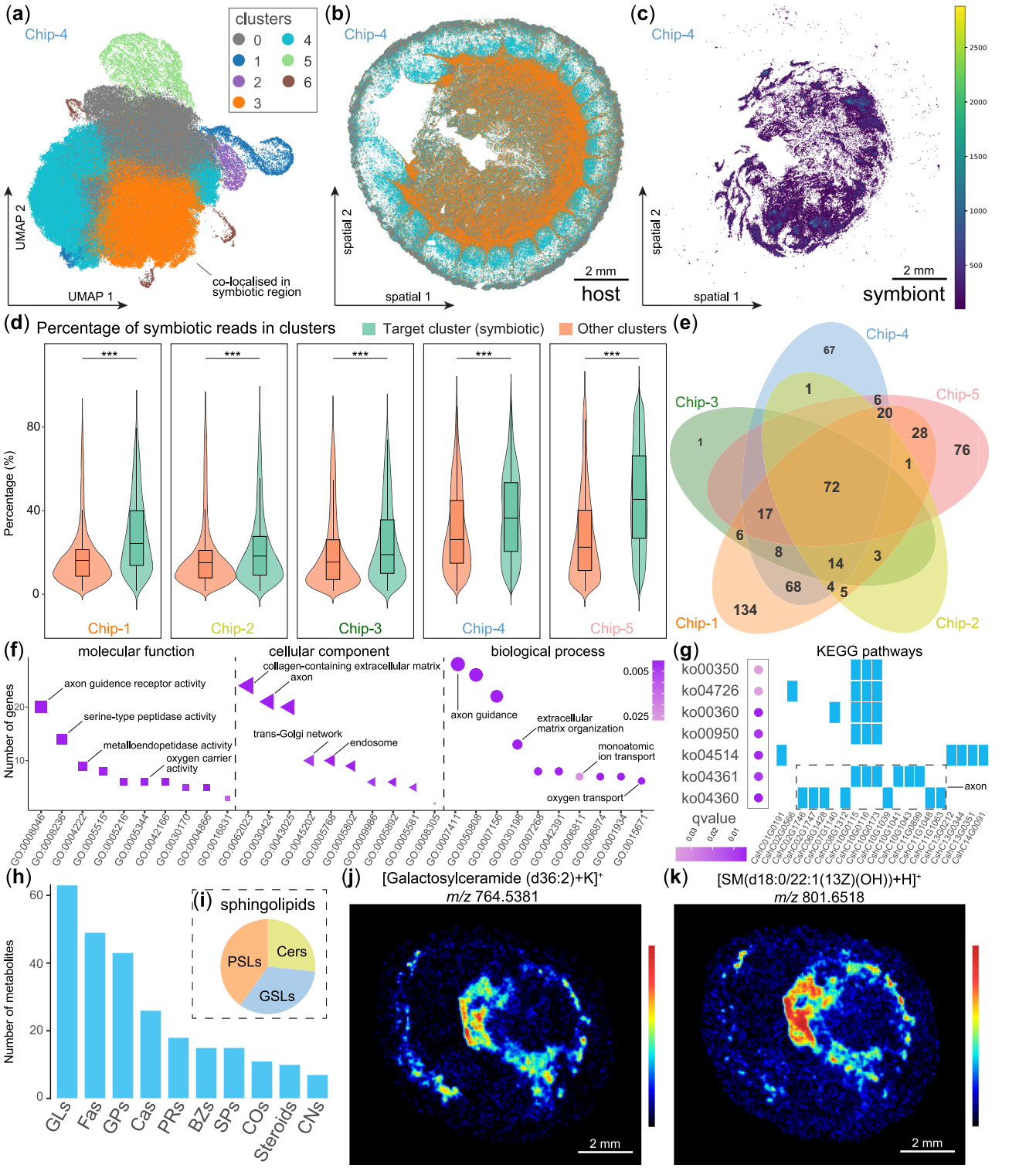
Our analysis identified molecules enriched in the symbiotic region, such as haemocyanins, proteases, and sphingolipids, which likely support the ongoing maintenance and controlled digestion of symbionts. However, defining the causal pathways for initial dracosphera formation and specific symbiont recognition is limited by our sample availability. We hypothesise that gene networks involved in phagocytosis and membrane remodelling drive early establishment, but directly testing these molecular mechanisms will require future studies. Particularly, juvenile specimens just after initial symbiont acquisition likely undergo intense formation of new dracosphera and will be helpful in revealing the initial pathways governing the formation of this symbiotic organelle.

Potential function of the dracosphera

Why would *C. shenloong* compartmentalise its symbionts into dracospherae when other chemosymbiotic organisms manage without such costly multilamellar organelles [8] ? One explanation lies in its adaptation to deep, anoxic, reducing environments. The habitat of *C. shenloong* is hypoxic, a typical challenge for vent and seep holobionts that must adapt to obtain both reducing chemicals and oxygen[48]. However, given its burrowing depth, *C. shenloong* may represent an extreme case even among chemosymbioses. Hypoxic and anoxic conditions can drive evolutionary innovations, such as the transformation of mitochondria into hydrogenosomes in loriciferans inhabiting permanently anoxic sediments up to 15 cm below the surface[49,50]. While *C. shenloong* likely extends its posterior gill to the surface for oxygenation, it also needs to burrow deep to take in hydrogen sulphide, which is only available at greater depths (Supplementary Fig. 1b). Consequently, *C. shenloong* may need to endure prolonged periods of anoxia.

Ultrastructure of the dracosphera superficially resembles other multilamellar elements known in nature, such as myelin sheaths around neural axons that both function as insulators and produce ATP by oxidative phosphorylation, as well as thylakoid membranes in cyanobacteria which are sites of photosynthesis[51]. A key function of these multilamellar elements is the absorption and release of gases, especially oxygen and carbon dioxide. Given the *C. shenloong* endosymbiont is microaerobic and requires a very low level of oxygen (and is potentially facultatively anaerobic), it is plausible that the host mollusc would benefit from preventing the symbiont from contacting

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oxygen; thereby retaining the available oxygen for use by the animal under hypoxic conditions. Furthermore, the *C. shenloong* symbiont may function more optimally under hypoxic conditions with its *cbb3* type cytochrome *c* oxidase, though the close relatives of the symbiont are not obligate anaerobes[36]. The multilamellar wall of dracospherae possibly functions as a barrier of oxygen diffusion, and at the same time conversely delivers carbon dioxide (Fig. 4b). The mitochondria were not uniformly distributed in the bacteriocytes but instead formed discrete aggregates within specific cytoplasmic regions (Supplementary Fig. 14), which partly supports our oxygen compartmentalisation hypothesis. The diazotroph *Frankia* symbiotic with angiosperm plants is known to use multilamellar external vesicles to defend against high oxygen concentrations since nitrogenase is oxygen-sensitive[52]. Many termites house anaerobic symbionts in their hindgut to break down lignin and cellulose, which the host accommodates by constructing an oxygen gradient inside the body[53]. It has also been speculated that many such structures have convergently evolved for energy production[51]. Nonetheless, due to the lack of living specimens available for experimentation, this hypothesis is primarily derived on correlative evidence and requires physiological validation. Future studies should

Fig. 3 | Insights from spatial transcriptomics and metabolomics. a Uniform manifold approximation and projection (UMAP) of clusters from aggregated bins (host) in chip-4, with the information of all five chips shown in Supplementary files Table 7. The information listed in Fig. 3d included data from three biological replicates. **b** Spatial representation of clusters (host) in chip-4. **c** The spatial representation of symbiotic reads in chip-4. **d** The percentage of symbiotic reads in symbiotic clusters and the others, shown in violin plots and box plots. Each observation is a 10-μm spatial-transcriptomic bin from one of five sections from three specimens, shown as symbiotic:others; these sections are not independent biological replicates. Exact *n* values are reported for the symbiotic and other-cluster groups in each chip: Chip 1, $n = 11{,}757{:}41{,}880$; Chip 2, $n = 7{,}283{:}28{,}219$; Chip 3, $n = 20{,}236{:}39{,}666$; Chip 4, $n = 120{,}504{:}193{,}771$; and Chip 5, $n = 30{,}823{:}67369$. Box plots show the median (centre line), the first and third quartiles (box bounds; 25th and 75th percentiles), as well as the minimum and maximum values (lower and upper whiskers). Statistical comparisons were performed using two-sided Wilcoxon rank-sum tests (Mann–Whitney U tests). *P* values were adjusted across the five chip-level comparisons using the Holm method. The exact adjusted *P* values were: Chip 1, $P = 3.669638 \times 10^{-885}$; Chip 2, $P = 1.190579 \times 10^{-69}$; Chip 3, $P = 2.233114 \times 10^{-306}$; Chip 4, $P = 4.312832 \times 10^{-1894}$; and Chip 5, $P = 2.668920 \times 10^{-2656}$, which were labelled as "***". **e** Venn plot of marker genes in symbiotic clusters from five independent analyses, with candidate genes labelled in bold if identified in more than 2 results. **f** Functional category of candidate genes based on the gene ontology. **g** Functional category of candidate genes based on the KEGG (Kyoto Encyclopaedia of Genes and Genomes) pathway. **h** Histogram plot of identified metabolites in DESI-MSI at the class level. **i** Pie plot of three categories of 15 identified metabolites affiliated to sphingolipids. **j** Spatial pattern of galactosylceramide (m/z 764.5381). **k** Spatial pattern of sphingomyelin (m/z 764.5381). GLs glycerolipids, Fas fatty acyls, GPs glycerophospholipids, Cas carboxylic acids and derivatives, PRs prenol lipids; BZs benzene and substituted derivatives, SMs sphingolipids, Cos organooxygen compounds, Steroids: steroids and steroid derivatives, CNs: organonitrogen compounds. Source data are provided as a Source Data file.

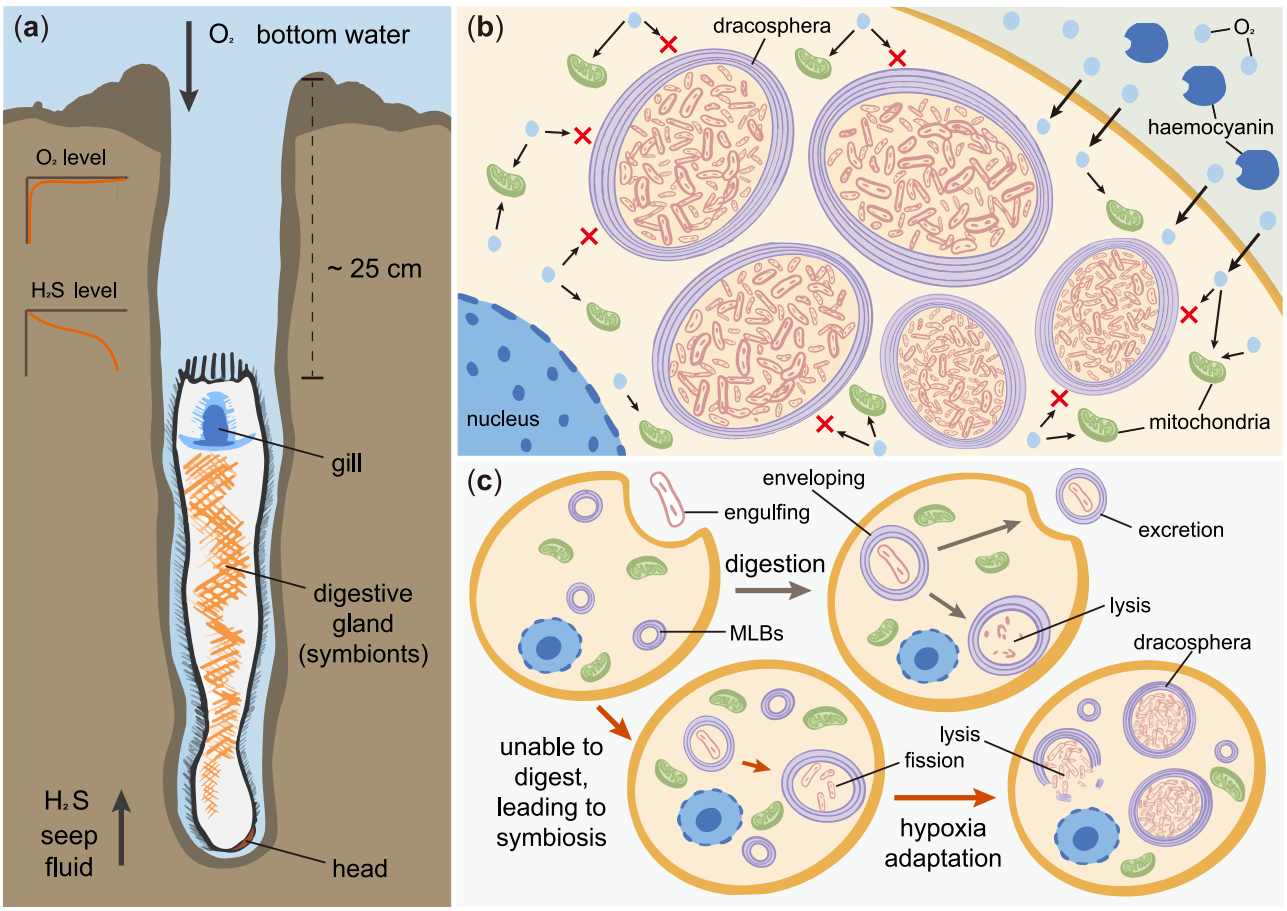


Fig. 4 | Schematic illustration of the chemosymbiosis in *Chaetoderma shenloong* using the specialised organelle, dracosphera. a Anatomy and life position of *C. shenloong* buried deep in anoxic seep sediment to take in hydrogen sulphide from the posterior gill. **b** Concept illustration for the inferred function of the dracosphera, where oxygen entering the host bacteriocyte cannot diffuse through to the symbionts due to the multilamellar wall of the dracosphera and is instead taken up by host mitochondria. **c** Proposed evolutionary pathway for dracosphera. Grey arrows indicate the likely original function of the multilamellar bodies (MLBs) where they are secreted to envelop and digest or excrete bacteria engulfed by the animal cell. Red arrows indicate the pathway where a sulphur-oxidising bacterium was able to resist lysis inside MLBs, and instead multiply inside them, eventually leading to a symbiosis with the host. With further adaptation to hypoxia, the membranes increased in layering to form the dracosphera to allow the host cell to utilise all available oxygen fully. These schematic illustrations were prepared by the authors with the assistance of Jinyi Chu (Ocean University of China) using Procreate, and used here with written permission.

aim to directly quantify the spatial oxygen gradient across the dracosphera and hypoxic tolerance of the living animal, utilising techniques such as microelectrode oxygen sensing or high-resolution oxygen-sensitive fluorescent imaging.

The structural similarity between myelin sheath and dracosphera is corroborated by the high expression of genes known to be linked with axon activities and the nerve system, such as fasciclin 2, semaphorins, and acetylcholine receptors (Supplementary Data 5) in the symbiotic organ. Having identified broad transcriptomic profiles, we investigated the spatial pattern of metabolites using desorption electrospray ionization mass spectrometry imaging (DESI-MSI, resolution: 50 μm). We found that 15 sphingolipids (Fig. 3h, i, Supplementary

Data 8 and 9) were enriched in the digestive gland. Among them, galactosylceramide (Galc) and sphingomyelin (SM) (Fig. 3j-k and Supplementary Fig. 15) are the major structural components in mammal nerve systems[54]. Ceramides, the precursors and products of these genes, also exhibited a similar pattern, indicating the active transformation among them (Supplementary Fig. 15 and Supplementary Data 5). Complementing these spatial metabolomic profiles, Raman spectra further corroborated the highly lipid-rich nature of the dracospherae compared to the ambient cytosolic fluid (Supplementary Fig. 16). There were higher expression levels of genes in membrane biosynthesis (e.g., serine palmitoyltransferase 1/2 and fatty acid synthase) in the symbiotic digestive gland tissue (based on bulk RNA-seq, Supplementary Fig. 12). Six haemocyanins were highly expressed in the digestive gland (oxygen carrier activity in Fig. 3f and Supplementary Data 5), which would be responsible for oxygen transportation.

Hypothesis for the evolution of dracosphera

The dracosphera could have arisen because of how the symbiotic relationship evolved. In the social amoeba *Dictyostelium discoideum*, facultative symbiosis with the bacteria *Burkholderia* is a by-product of the amoeba packaging bacterial cells with multilamellar membranes for digestion[55]. *Burkholderia* can resist digestion because the multilamellar envelope helps it tolerate harsher conditions[56]. We speculate that the symbiosis between *C. shenloong* and its symbiont may have originated in a comparable way (Fig. 4c), with the dracosphera protecting the bacteria and the bacteria providing energy in return. Though genes related to membrane biosynthesis were not expanded, there were higher expression levels in the digestive gland. This localised upregulation is likely dedicated to the active production of multilamellar membranes for the dracosphera, indicative of host transcriptional regulation (Supplementary Fig. 11), further supported by the high metabolic demands of chemosynthetic primary production.

Due to its unusual structural integration with symbiotic bacteria, it is important to clarify the dracosphera's position on the symbiosis continuum of organelles accurately. By definition, an organelle is a specialised, structurally distinct intracellular compartment dedicated to a specific physiological function, such as the phosphate-sensing organelle in *Drosophila melanogaster*[57]. Under this functional paradigm, the dracosphera qualifies as a chemosynthetic organelle, similar to the kleptosome in the photosynthetic sea slug *Elysia crispata* which stores the stolen chloroplasts from algae[58]. However, its integration is different from classical endosymbiotically-derived organelles such as mitochondria or plastids. Mitochondria represent the endpoint of organellogenesis from the symbiont itself, characterised by irreversible evolutionary events such as high endosymbiont genome reduction, obligate vertical inheritance, and host-to-symbiont protein import machineries. The recently discovered nitroplast in the marine algae *Braarudosphaera bigelowii* derived from an N_2-fixing endosymbiont *Candidatus* Atelocyanobacterium thalassa has been integrated into the algal cell architecture including cell division, similar to mitochondria[3]. In contrast, the dracosphera appears to represent an alternative route like the kleptosome where the host has produced an organelle to house the symbiont. It maintains a dedicated structural interface and is governed by localised host transcriptional regulation, yet currently it lacks evidence for profound genomic integration and inheritance with cell division. How newly divided host bacteriocytes acquire and maintain both the dracosphera and the thioautotrophic symbionts remains to be seen, and requires experimental studies with live animals. It is possible that membrane-bound structures such as the dracosphera and kleptosome[58] paves the way to genomic integration between the two parties towards the level of mitochondrial and chroloplast.

As multilamellar bodies can now be synthesised using artificial cell techniques[59], this mode of establishing symbiosis could be tested in the future via experimental enveloping of bacterial lineages. Though chemosynthesis is ubiquitous in marine realms, the dracosphera is the only known organelle dedicated to it and presents an opportunity to explore organellegenesis, pending future studies on its origin and machinery. Characterising the components and the biochemical composition of the dracosphera multi-membrane by isolation and analyses with methods such as mass spectrometry will also help us to better understand its origins and functions.

Potential global distribution and impact of chemosymbiotic aplacophorans

Our discovery of a adaptation from one of the most poorly understood animal groups highlight how little we know about the potential of marine invertebrates, especially infaunal deep-sea taxa[60]. During this study, we serendipitously collected an individual of *C. shenloong* from a seep site near a mud volcano ('MV3') off Kyushu in Japan (more than 98.89% ANI at the whole genome level, Supplementary Fig. 10). This extends the range of this species from two seeps in the South China Sea (Haima and Jiaolong Ridge[21,22] to Japan. Despite its enormous size and unusual ecology, *C. shenloong* has remained undiscovered across its wide range until recently. The dracosphera enables aplacophorans, and other animals to live much deeper in anoxic mud than previously thought. Previously, all molluscan chemosymbioses have been from conchiferans, but *C. shenloong* is an aculiferan caudofoveate. Given that the largest known living caudofoveate (*C. felderi* reaching 40 cm length) resides in cold seeps in the Gulf of Mexico[21,61], chemosymbiotic aplacophorans may be much more common than we realise and contribute significantly to primary production at seeps and the deep ocean as a whole[7].

Chemosymbiotic animals exhibit unusually high growth rates and are the dominant sites of biomass production at cold seeps[62]. Our understanding of the biogeochemical cycling and carbon budget at these habitats so far[63,64] have largely been based on the biodiversity and biomass of species visible on the surface, as infaunal holobionts are typically rarer at seeps. The discovery of *C. shenloong* as the first representative of a likely widespread group of deep infaunal chemosymbiotic animals shifts this perception. The isolation of symbionts in the dracosphera allows these giant aplacophrans to efficiently utilise the limited oxygen and access sulphides deep in the sediment to fix carbon. This adaptation allows these holobionts to thrive inside anoxic sediments where other animals cannot. Given their body sizes and apparent high densities, 'dark carbon fixation' by these aplacophorans invisible from the surface may account for a high proportion of the total carbon budget at seeps. Meanwhile, the highly efficient sulphur cycling in the dracosphera consumes sulphides before they diffuse upwards into the water above the sediment layer. This activity likely alters the geochemical gradient, potentially reducing sulphide toxicity for shallower infauna and changing the resource availability for surface-dwelling chemosynthetic communities. Furthermore, burrowing by large megafauna like *C. shenloong* would also modify the sediment environment by pumping the chemical gradient[64], especially as the animal is expected to move up and down the burrow. Symbiotic aplacophorans, powered by dracosphera, therefore could represent an important but overlooked contributor to biogeochemical cycling at cold seeps worldwide. Many more such evolutionary marvels, with potential significance in geochemical cycling, could still be hidden in the deep sea.

Methods

Animal-ethics statement

This study involved field-collected marine invertebrates and did not involve vertebrate animals or other organisms requiring institutional animal-ethics review under the applicable regulations. All collection activities were conducted in compliance with applicable local,

national, and international regulations; no collection permits were required for sample collection.

Sample collection and preservation

Eight individuals of *Chaetoderma shenloong* were collected from Haima methane seep in the South China Sea using a 50 cm long push-core mounted on the remotely operated vehicle (ROV) *Pioneer* on the research vessel (R/V) *Xiangyanghong 01* of the First Institute of Oceanography, Ministry of Natural Resources (MNR), China, September 2022. One other individual was collected from a cold seep associated with the MV3 mud volcano off Tanegashima, Japan by an Okean grab on-board R/V *Hakuho Maru* cruise KH-23-4 Leg 2 (chief scientist: Akira Ijiri, Kobe University)[65]. For detailed collecting data see Supplementary Table 1. The H_2S, O_2, and nitrate (NO_3^-) concentrations in the sampling pushcore were measured from the sediment samples using Unisense H_2S and O_2 microsensors attached to a micromanipulator. The raw concentrations of them were shown in Supplementary Table 2-4 and Supplementary Fig. 1.

Samples were preserved in multiple ways after recovery on-board the research vessels, using the following methods: (1) flash-frozen in liquid nitrogen in 15 ml tubes and then stored at −80 °C; (2) specimens were fixed in RNAlater (ThermoFisher, USA) and then stored at −80 °C; (3) fixed in 4% paraformaldehyde fix solution (PFA) overnight, then substituted by 100% methanol and stored at −20 °C; (4) fixed and stored in 2.5% glutaraldehyde solution at 4 °C; 5) fixed and stored in 95% ethanol at room temperature. The different analyses are shown in Supplementary Table 5.

Stable isotopic analyses

The digestive gland (hosting symbionts) and muscle (non-symbiotic tissue) from four specimens were dissected and freeze-dried. The ground powders (~ 0.1 g each) were subjected to an element analyser Flash 1112 (ThermoFisher) for the determination of carbon and nitrogen.

μ-CT scanning and 3D reconstruction

The specimen fixed by PFA and transferred to methanol was subjected to a rehydration series with a decreasing concentration of methanol to Phosphate Buffered Saline (PBS) solution (90%, 80%, 70%, 60%, 40%, 20%) for 1 h each, and then placed into pure PBS for 24 h. The spicules were first roughly removed with scalpels and needles, and then placed in 0.1 M HCl solution until complete decalcification as indicated by a lack of emerging bubbles. The specimen was then stained with 3% Lugol's solution for two days.

A SkyScan 1276 μ-CT scanner (Bruker, Belgium) was used to scan the specimen, placed in a 50 ml centrifuge tube filled with water. The scan was conducted at 55 kV and 200 μA, at a pixel resolution of 6.54 μm. The software NRecon (Bruker, Belgium) was used to obtain the 2D sections based on the scanned X-ray images. These sections were processed with CTAn and DataViewer software (Bruker, Belgium). The specialist software Amira v2022.2 (Thermo Fisher Scientific) was used for 3D reconstruction of the internal organ, by manually highlighting the organs in each section and then generating the 3D surface.

Scanning electron microscopy

A specimen preserved in 2.5% glutaraldehyde solution was cut into small pieces (about 1 ~ 2 mm thickness) and then washed in 0.1 M PBS for 15 min, repeated three times. The specimen was cut into small pieces 1–2 mm in thickness, and then critical-point dried. The dry specimens were sputter-coated with gold (Au), and then imaged using a VEGA3 SEM (TESCAN) at 10.0 kV.

Transmission electron microscopy

Tissue from a specimen fixed in 2.5% glutaraldehyde solution was cut and washed in the same manner as for SEM observations. Subsequently, the tissue was fixed in 1% osmium tetroxide (OsO_4) for 1 h and then washed in 0.1 M PBS for 15 min three times. The specimen was dehydrated in an increasing methanol series (50, 70, 90, and 100%, three times each) for 15 min each and embedded in resin (Epon 812). Thin sections at 70 nm thickness were cut using a Reichert ULTRACUT Ultrathin slicer (Leica, Austria). The thin sections were stained using the uranyl acetate and lead citrate double staining method[66], with a uranyl acetate and lead citrate staining for 15 min each. Images were captured using a JEM 1200-EX (JEOL, Japan) TEM at an accelerating voltage of 80 kV.

Focused-ion beam scanning electron microscopy

We used a focused-ion beam scanning electron microscope (FIB-SEM) to construct the 3D structure of the dracosphera. The specimen was stained using the following procedure: washed in 0.1 M pH 7.2 phosphate buffer (PB) for 3 min for three times; stained in 0.1 M PB containing 1.5% ferrocyanide and 2% OsO_4 for 1 h under dark environment and on the ice then washed in double-stilled water for 3 min for five times; stained again in 1% thiocarbohydrazide (TCH) solution at room temperature for 20 min followed by washing in double-stilled water for 3 min for five times; stained in 2% OsO_4 in water for 30 min under dark environment and on the ice then washed in double-stilled water for 3 min for five times; stained in 1–2% uranyl acetate in water at room temperature for 30 min then washed in double-stilled water for 3 min for four times and again at 60 °C oven for 3 min; and finally in Walton's Lead (39.9 mg L-aspartic acid and 66 mg lead nitrate in 10 ml double-stilled water, pH=5.5) at 60 °C oven for 3 min and then washed in 60 °C double-stilled water for 3 min for five times.

The stained specimen was dehydrated in an increasing serial ethanol solution (30%, 50%, 70%, 80%, 90%, 95%), twice in each solution for 5 min, then twice in 100% ethanol for 10 min. Resin embedding was done in the following steps: 5 mins in 1:1 ethanol/acetone solution, 1 h in resin/acetone solution (1:2) at room

perature followed by 1 h in 1:1 resin/acetone solution and then 1 h in 2:1 resin/acetone solution (2:1), and finally in pure resin at room temperature for 1 h, and repeated three times. The resin-embedded specimen was hardened in a 60 °C oven for 2 days. A Crossbeam350 (ZEISS, Germany) FIB-SEM was used to image the specimen at a resolution of 10 nm.

Metagenome sequencing and assembly

Genomic DNA was extracted from the symbiotic digestive gland using the SDS method and then randomly broken into about 350 bp fragments. After library construction, it was sequenced with the paired-end 150 bp mode on an Illumina Novaseq 6000 platform. The raw reads were trimmed in trimmomatic version 0.39[67] with the following settings (TruSeq3-PE-2.fa:2:30:10:8:true SLIDINGWINDOW:5:20 LEADING:3 TRAILING:3 MINLEN:36). MEGAHIT version 1.2.9[68] was used in metagenome assembly with the default settings. MaxBin2 verion 2.2.7[69] was used to retrieve the symbiont genome and the fragmented host genome. We also downloaded and reanalysed reads from a previous work on *Chaetoderma shenloong*[22].

Genome sequencing and assembly

Muscle and gill tissues from a flash-frozen individual were used in high-molecular-weight DNA extraction using the SDS method. The quality-checked DNA was sent to Novogene Co. Ltd (Tianjin, China) for library construction and sequencing. High fidelity (HiFi) sequencing was adopted in the genome assembly, with the platform and outputs shown in Supplementary Table 6. Meanwhile, a part of the tissue was processed in high-throughput chromosome conformation capture (Hi-C) library with the restriction emzyme MBoI as the guideline, and the PE-150 reads were output from an Illumina Novaseq 6000 platform.

Reads were extracted from.bam file using extracthifi version 1.0.0 and then the adapters were checked using HiFiAdapterFilt version 2.0.

Before assembly, minimap2 version 2.24-r1122[70] was employed in mapping HiFi reads to the draft genome of the symbiont (from the NGS-based metagenome). Therefore, HiFi reads were divided into the symbiont reads and the symbiont-free reads. The symbiont HiFi reads were subjected to hifiasm_meta version 0.3-r063.2[71], which successfully assembled the complete genome of symbiont. The symbiont-free reads were subjected to hifiasm version 0.16.1-r375[72] for genome assembly with the default settings. The duplicated and contaminated contigs in the primary assembly were identified using minimap2, blastn[73], Purge_Dups version 1.2.5[74], TaxonKit version 0.15.1[75], diamond blastp version 2.0.15[76], and MEGAN version 6.21.7[77], which resulted in the non-redundant and decontaminated contigs of *C. shenloong*.

To anchor these contigs to the superscaffold level or pseudo-chromosome level, the Hi-C reads were processed in HiC-Pro version 3.1.0[78] to remove the unlinked reads. Then, they were subjected to Juicer version 1.6[79] for the output of merged_nodups.txt, and then the contigs were subjected to the 3D de novo assembly (3D-DNA) pipeline version 201008[80]. Juicebox Assembly Tools version 2.17.00 (Dudchenko et al. 2018) was employed to generate and visualize a Hi-C contact map, and then the mis-assembly or mis-scaffolding was manually corrected. Lastly, the reviewed links were subjected to 3D-DNA for constructing the pseudo-chromosome-level genomes. The completeness of the genome, transcriptome, and the proteome (see below) was evaluated using Benchmarking Universal Single-Copy Orthologs (BUSCO) version 5.2.2[81] against the metazoan odb10.

Repeat region and gene model prediction of the host genome

The general pipeline was slightly modified from a published work[82]. Briefly, the soft-masked genome and the corresponding repeat level were called using RepeatModeler v2.0.3[83] and RepeatMasker v4.1.2-p1[84]. To obtain the transcriptional profile, the RNALater-preserved sample was dissected into different organs (gill, muscle, digestive gland, pericardium, gonad, and epidermis) and then used in RNA-seq. In trimming metagenomic reads, the low-quality or adapter-contaminated RNA sequences were removed using trimmomatic. These reads were then aligned to the soft-masked genome using aligner STAR version 2.7.10a[85], which generated.bam files according to mapping regions. BRAKE2 version 2.1.6[86], implemented with Augustus version 3.4.0 and GeneMark version 3.67_lic for the ab inito gene prediction, was employed in ab-initio prediction ("augustus.gff3" and "genemark.gff3"). The protein-coding regions were also predicted via homologue alignment, with proteins from 28 high-quality genomes in Metazoa aligned to the soft-masked genome using miniport version 0.5-r179[87] ("proteins.gff3"). Trinity version 2.13.2 was adopted in assembling transcripts with both de novo and genome-guided modes, whose outputs were aligned to the genome using PASA version 2.5.2[88]. Meanwhile, Stringtie v2.1.1[89] was also used in predicting the expressed regions in genome, which was used in Evidencemodeler version 1.1.1[88] to integrate the above three predictions into a comprehensive profile of genes, and the weights of evidence were AUGUSTUS for 2, GeneMark for 2, PASA for 10, Stringtie for 6, and homologue from protein for 5. The EVM consensus predictions were compared with the assembled transcripts and gene models using PASA[88], which also led to the identification of untranslated regions (UTRs) and alternatively spliced isoforms in genes. The gene functions were predicted against Nr, and SwissProt, and Kyoto Encyclopaedia of Genes and Genomes (KEGG) using diamond blastp[76]. The information of gene ontology was retrieved using blast2go version 5 basic.

Comparative genomics and gene quantification

The phylogenetic relationship among chemosynthetic hosts was investigated using VEHoP version 1.0 (IQ-TREE2, -m MFP -B 1000). The full datasets of genomes involved in phylogenomics are shown in Supplementary Data 1. The divergent time among nodes were predicted using mcmctree embedded in paml version 4.10.7[90], using fossil records for calibration (the code used is included in Supplementary Code 1, while the details of the calibration points are shown in Supplementary Data 10). Then, the changes of gene family were calculated using CAFE version 5[91]. The expression level of genes was quantified using STAR for RNA-seq reads mapping and Stringtie for reads count evaluating.

Symbiont genome processing

We sequenced the symbiont-hosting digestive gland from nine individuals of *C. shenloong*. For intact specimens, two samples were taken, one from the anterior and one from the posterior. The symbiont genomes were obtained from a total of 11 samples, including a complete-level one from HiFi and fragmented ones from metagenomic short reads. The published symbiont genomes in Mollusca and Annelida were shown in Supplementary Data 2. To avoid annotation bias, we repeated the annotation and evaluation of the above-mentioned genomes using Bakta version 1.9.2[92] and CheckM2 version 1.0.1[93]. The ortholog information of proteins in Kyoto Encyclopaedia of Genes and Genomes (KEGG) was retrieved using diamond blastp against locally deployed database (classification: prokaryotes). FastANI version 1.32[94] was employed to compare the similarity among these symbiont genomes at the nucleotide level. The taxonomy of symbionts genomes was at first classified using GTDB-Tk version 2.3.2[95] and then further checked using VEHoP version 1.0[96] at the phylogenomic level.

We also used population genetics to study the genetic connectivity of both the host and the symbiont, as in previous works[43,97], in order to obtain insights into the symbiont. In short, the metagenomic reads were aligned into the *C. shenloong* symbiont using bowtie2 version 2.5.1[98] and then the single nucleotide polymorphisms (SNPs) in each sample were called and extracted using GATK version 4.5.0[99].

Host genetic distance and similarity among specimens

The metagenomic sequencing reads were assembled into contig-level genomes using MEGAHIT version 1.2.9[68] as described above. The mitochondrial genomes of each specimen were extracted using the blastn of the COX1 (cytochrome oxidase *c* subunit I) gene against the fragmented contig-level genomes. FastANI version 1.32[94] was further employed in quantifying the similar levels regarding to the mitochondria. Meanwhile, the similarity at the whole-genome level was also checked using skani version 0.3.1[100] based on all draft assemblies from MEGAHIT. The generated matrix was visualised using TB-tools II version 2.309[101].

Spatial transcriptomics

We used Stereo-seq FFPE (BGI, China) to investigate the spatial heterogeneity of transcripts and to investigate the interactions between the two symbiotic parties. The newly developed kit (Stereo-seq FFPE kit 201SN114, BGI) enables us to detect transcripts simultaneously from both eukaryotic and prokaryotic sources, which empowered us to visualize the spatial pattern of symbiotic interactions. Briefly, this method adopts random probes with coordinate IDs (CIDs) to capture the total RNA in a section and then processed in complementary DNA (cDNA) reverse transcription.

The *C. shenloong* posterior part containing both the symbiotic digestive gland and the gonad, was immersed in 4% PFA overnight and then immersed in 100% methanol. Tissues were dehydrated in PBS/methanol solutions (3:1, 1:1, and 1:3, each concentration for 30 min) and permeabilised in 100% ethanol for 15 min, 1:1 xylene ethanol solution for 15 min, and in xylene twice for 10 min each. Then, the tissue block was embedded into paraffin after short immersion in a serial xylene paraffin (1:1 for 10 min and 100% paraffin for 20 min). Five serial sections at 5 μm thick (labelled as No. 1 to No. 5) were cut. The middle section was adhered to the surface of the Stereo-seq chip (BGI, China)[102]. The other four sections were mounted on coated slides and used in downstream FISH or HE staining. The chip was incubated in a

thermocycler at 42 °C for 3 h, 37 °C overnight, and then 60 °C for 1 h. Then, the chip with section was placed in a set of solutions at room temperature for the removal of paraffin, as follows: Histo-clear solution 1 for 20 min; Histo-clear solution 2 for 20 min; 100% ethanol for 5 min; 100% ethanol (renewed) for 5 min; 96% ethanol for 5 min; 96% ethanol (renewed) for 5 min; 90% ethanol for 2 min; 80% ethanol for 2 min; 70% ethanol for 2 min; 50% ethanol for 2 min; 30% ethanol for 2 min; and finally double-stilled H_2O for 1 min. The chip was then air-dried in a fume hood.

The dried chip was transferred to ssDNA staining (ssDNA for Qubit) and fluorescent imaging with a Motic Custom PA53 FS6 microscope prior to in situ capture at the channel of FITC. Subsequently, the stained ssDNA in the chip was de-crosslinked in FFPE Decrosslinking Reagent at 95 °C for 30 min. After fixing with pre-chilled methanol (-20 °C) for 20 min, the section on the slide was digested in a permeabilisation solution (10 mM Tris-HCl, 25 mM EDTA, 100 mM NaCl, 0.5% SDS) at 37 °C for 30 min and then washed with 5% RNase inhibitor in 0.1x SSC buffer (Thermo, AM9770). The cDNA was reverse transcribed in the chip with the FFPE MIX solution (158 µL FFPE RT Buffer mix, 30 µL FFPE RT Enzyme mix, 10 µL FFPE RT Oligo, 2 µL FFPE Dimer) at 42 °C for 5 h. The cDNA-containing chips were then subjected to Prepare cDNA Release Mix (cDNA Release Enzyme, cDNA Release buffer) treatment, overnight at 55°C. The harvested cDNA was purified using the VAHTSTM DNA Clean Beads (0.8×) and then amplified in the amplification solution (42 µL cDNA, 50 µL cDNA Amplification MiX, and 8 µL FFPE cDNA Primer Mix). The PCR protocol was: 95 °C for 5 min, 15 runs of [98 °C for 20 s, 58°C for 20 s, and 72 °C for 3 min], and 72 °C for 5 min. After quantification using the Qubit dsDNA HS kit, the cDNA product was used in library construction under the guidelines of Stereo-seq 16 Barcode library kit V1.0.

Fastq files were generated using an MGI DNBSEQ-T7 sequencer. The whole procedure was processed using a publicly available pipeline SAW available at https://github.com/BGIResearch/SAW. Briefly, The genome index was built with the.gtf and.fasta files using SAW makeRef. CID and MID are contained in 'read 1' (CID: 1-25 bp, MID: 26-35 bp) while 'read 2' consists of the cDNA sequences. The CID sequences on the first reads were first mapped to the designed coordinates of the in situ captured chip achieved from the first round of sequencing, allowing 1 base mismatch to correct for sequencing and PCR errors. Reads with MID containing either N bases or more than 2 bases with a quality score lower than 10 were filtered out. The CID and MID associated with each read were appended to each read header. Retained reads were then aligned to the reference genome using STAR and mapped reads with MAPQ > 10 was counted and annotated to their corresponding genes. UMI with the same CID and the same gene locus were collapsed, with the criteria of a maximum of 1 base mismatch. Finally, this information was used to generate a CID-containing expression profile matrix. To analysis the cell-level expression, given the size of 'dracosphera' and eukaryotic cell (< 20 µm), these spots were aggregated into bins (bin 20, 20 ×20 DNA Nano Balls per bin, approximately 10 µm in diameter). To lay to the gene expression pattern in the symbiotic region, the unsupervised clustering of bins was performed at a resolution ranging from 0.2 to 0.4 until the match between most bins in a single cluster and the symbiotic region. The marker genes in clusters were identified with the threshold below: log2fold_change > 1 and adjusted *p*value < 0.001. The functional enrichment of candidate genes were processed using ClusterProfiler version 4.2.0[103].

Spatial metabolomics

The flash-frozen samples were used in analysing the spatial patterns of metabolites in *C. shenloong*, focusing on the symbiotic digestive gland and the interactions between symbiont and host. A whole individual of *C. shenloong* was divided into 10 parts from anterior (No.1) to posterior (No. 10) on dry ice while still frozen. Among them, three parts with the symbiotic organ were selected to represent the different sections, including No.2 (around the neck), No.4 (middle of trunk), and No.8 (close to the posterior). After mild thawing, these tissue parts were slowly placed into the optimal cutting temperature (OCT) compound (Cat. 4583, SAKURA) in the mould while carefully avoiding bubbles. Then, the moulds were quickly frozen with the help of dry-ice and transferred to a freezer at -80 °C until ready for sectioning. Before tissue sectioning using a cryostat (CM1950, Leica), the embedded tissue blocks were moved to a cryostat chamber for 20 min for the equilibration of temperature. The thickness of the section was set as 20 µm and the sections were mounted onto coated slides. In each tissue block, we harvested a set of three serial sections. Slides were stored and transported at -80 °C. Among the three serial sections, the middle section was subjected to SELECT SERIES MRT (Waters, USA) for desorption electrospray ionisation - mass spectrometry imaging (DESI-MSI), and the other two were used for FISH or HE staining.

The DESI-MSI analysis was performed on a SELECT SERIES@ MRT high mass resolution mass spectrometer (Waters) equipped with DESI XS source. General acquisition parameters were: positive and negative; MS acquisition rate: 0.5 s; source sprayer voltage: 0.5 kV; sampling cone voltage: 50 V; source temperature: 120 °C; source sprayer gas pressure: 0.08 Mpa; DESI solvent: 98:2 methanol/water; DESI solvent flow rate: 1.8 µl/min. Pixel sizes: 50 µm. Data was acquired using Masslynx software version 4.2. The.raw outputs were transformed into.imzML files using HDI version 1.6. Then, the results from four scans in the negative mode and five scans in the positive mode were imported into Cardinal for processing, including the background removal, peak alignment, and peak selection. The metabolites were annotated against *LuMet-Space version 1.0* using PySM and mass2adduct strategies. The spatial patterns of metabolites in *C. shenloong* were investigated using spatial shrunken centroids clustering (sscc), T-Distributed Stochastic Neighbour Embedding (TSNE), and Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP). The resolution here was 50 µm, equal to the step size of a DESI solvent sprayer. This was the highest resolution in our data with spatial information, although this was still larger than the cell size (10-20 µm). Due to different specimens being used in spatial transcriptomics and spatial metabolomics, we could not directly compare the same sections across the two methods. Therefore, we adopted the most appropriate resolution in each analysis, 50 µm in spatial metabolomics and 10 µm in spatial transcriptomics, respectively.

Hematoxylin and eosin (H&E) and fluorescence in situ *hybridisation (FISH) staining*

Staining for H&E and FISH was done using OCT-embedded and paraffin-embedded sections. The pre-staining steps differed according to embedding methods, with paraffin removal for the paraffin-embedded sections and fixation by 4% PFA at room temperature for 10 min for OCT-embedded sections.

H&E staining was carried out using the following protocol. The nuclei were stained with haematoxylin (G1004, Servicebio, China) for 3–5 min followed by a rinse in distilled water; section was differentiated with 1% HCl in ethanol for 2–5 sec (G1039, Servicebio, China) followed by another rinse in distilled water; slide was stained again with the Dako Bluing Buffer (G1040, Servicebio, China) and rinsed in distilled water; subjected to a dehydration series in an increasing concentration of ethanol (70%, 80%, 95%, and 100%, 3 min in each one); and finally stained with eosin (G1002, Servicebio, China) for 1–2 min followed by dehydrating in 100% ethanol for 5 mins (repeated three times). The slides were with xylene for 5 min.

Our FISH procedure followed a previously published protocol[104]. The specific probe targeting the 16S region of the *C. shenloong* symbiont was designed using the online platform Primer-blast[105], named Csh_SOB (5′-CGTCAAGGTTAGTAGGTATTAGCTACTAGC-3′) and labelled with Sulfo-Cyanine3 (cy3, Excitation: 550 nm, Emission: 570 nm).

Furthermore, the universal probe targeting all bacteria EUB338 (5′-GCTGCCTCCCGTAGGAGT-3′) was also adopted in verifying the spatial location of symbionts in the tissue, labelled with Sulfo-Cyanine5 (cy5, Excitation: 633 nm, Emission: 670 nm). The two probes were synthesised by Sangon Biotech (China). Two more dyes, DAPI (Cat. C0065, Solarbio, China, Ex: 364 nm, Em: 454 nm) and CellMask™ (Cat. C37608, ThermoFisher Scientific, Excitation: 522, Emission: 535 nm), were used to localise dsDNA and membrane, respectively. MitoTracker (Cat. M7514, ThermoFisher Scientific, Excitation: 490 nm, Emission: 516 nm) was employed to visualise the distribution of mitochondria in sections. Hybridisation of the bacterial probes was processed in a buffer (5 μg/mL probe in 0.9 M NaCl, 0.02 M Tris-HCl, 0.01% sodium dodecyl sulphate and 30% Formamide) at 46 °C for 1 h. Then, they were washed in a washing buffer (0.1 M NaCl, 0.02 M Tris-HCl, 0.01% sodium dodecyl sulphate, and 5 mM EDTA) at 48 °C for 10 min. Subsequently, sections were co-stained with DAPI (10 μg/mL) and CellMask for 10 min. After washing using PBST (Tween-20: PBS = 1:1000), the slides were mounted with ProLong Diamond Antifade Mountant (Invitrogen). The images were captured in the confocal models LSM900 (CZISS, Germany) and Dragonfly 200 (Andor), with 405 nm for DAPI, 488 nm for CellMask and MitoTracker, 561 nm for symbiont, and 647 nm for EUB338. A total of 28 sections were processed from three independently prepared specimens. All three specimens showed the similar patterns to those presented in Fig. 1g.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The raw sequencing reads in this work have been deposited in the NCBI database as a BioProject, with the genome assembly (i.e., HiFi, Hi-C, RNA-seq data) in PRJNA1149698 [https://www.ncbi.nlm.nih.gov/bioproject/1149698], genome resequencing reads in PRJNA1206971 [https://www.ncbi.nlm.nih.gov/bioproject/1206971], and spatial transcriptomics data in PRJNA1206974 [https://www.ncbi.nlm.nih.gov/bioproject/1206974]. The mass spectra in spatial metabolomics were deposited in the EMBL-EBI MetaboLights database under the accession number MTBLS15431. The assembled genome is available in NCBI GenBank (accession number: JBLUWO000000000.1 [https://www.ncbi.nlm.nih.gov/nuccore/JBLUWO000000000]); the host and symbiont gene feature files are also available on Figshare [https://doi.org/10.6084/m9.figshare.28143302]. The raw μ-CT data from which the 3D anatomical reconstructions were completed are available also on Figshare [https://doi.org/10.6084/m9.figshare.32615439]. The processed files recording gene expression and clusters are available on Figshare [https://doi.org/10.6084/m9.figshare.33335040]. The previously available genomic data used in this study are shown in Supplementary Data 1 and 2 for animals and microbes, respectively. Source data are provided with this paper.

Code availability

The commands used in this study can be found in Supplementary Code 1.

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Competing interests

The authors declare no competing interests.

Additional information

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